

Recall from Chapter 3 that the concept of correlation coefficients measures the extent to which two factors move in the same or opposite direction. In the Table 6.2 set of results, first notice how all the correlation coefficients in the table are negative. As you might expect, this means that if you experience worse-than-average portfolio investment returns at any time during your retirement, your income sustainability will be lower than expected. The same goes for increased longevity. If you live longer, you must spend more and experience higher inflation during retirement. They are all risk factors and will have a negative impact. Think of them as the factors that reduce your retirement sustainability.

Notice, for example, that the correlation coefficient for the first seven years is -56.3% while the correlation for the second seven years is only -27.5% . The way to interpret this number is as follows: In the many simulations we generated, some resulted in better-than-expected sustainability and others were worse. In fact, half of the time things were better than expected and half of the time they were obviously worse. Of course, we are definitely not advocating that a 50% probability of financial retirement success is an acceptable Monte Carlo number (it isn't), but rather we are using this neutral 50/50 simulation as the basis for the correlation analysis.

In the cases for which the first seven years (from age 65 to age 72) were lousy, more often than not (roughly 56% of the time) the desired income was not sustainable, and the retiree had to reduce his standard of living. However, if the second seven year's return is lousy (between the 73rd birthday and the 79th birthday), the impact on sustainability is lower: 27% of the time, the income was not sustainable.

Just as interesting is the impact of longevity risk. I return to this important topic in the next chapter, but for now think of longevity risk as the risk that you outlive all of your human and financial capital. Notice that its correlation coefficient is -53.9% in Table 6.2, which is almost as high and important as the impact of the first seven years with a coefficient of -56.3% . In other words, if someone were to ask me, "What has a worse impact on income sustainability? Is it underestimating my life expectancy or getting unlucky in the first few years of retirement?" My answer would be that they are roughly on the same order of magnitude. Be fearful of them both.